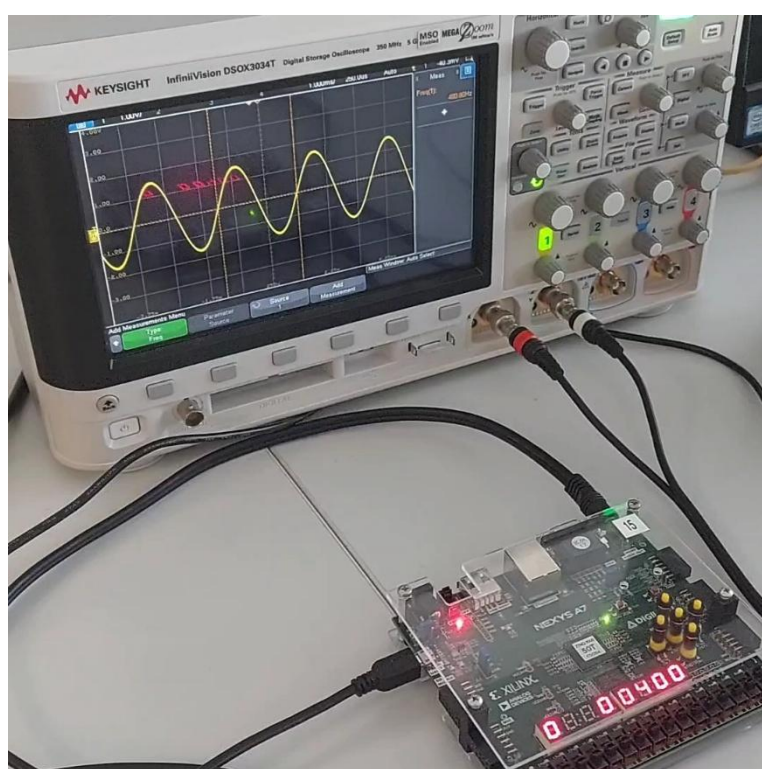


Introduction:

This project focuses on creating a functional waveform generator implemented on the Nexys A7-50T board. The generator is capable of switching between four different waveforms: harmonic, triangular, square and sawtooth, as well as changing frequency. The frequency can be set in range from 1 Hz to 10 kHz and is treated so that the user cannot exceed these limits. Recommended functional frequency range is from 50 Hz to 5 kHz due to distortion caused by integrated internal filter. User interface consists of four buttons for selection and modification of parameters, one button for reset into default state and a switch which acts as an executor allowing or shutting down the output. Current parameter settings are displayed on seven-segments and the parameter selected for modification is indicated by blinking of the segment.



Outputs:

Seven-segment display:

Segment 0-4 – display of the set frequency
Segment 7 – display of the selected waveform

Audio output:

Pulse width modulated signal is outputted at 3.5 mm jack connector

Architecture:

Button debounce is modified to generate periodic pulses when held.

Parameter block serves as the core logical block which outputs selected waveform, frequency and information to be displayed. Frequency is then converted into corresponding phase increments. The phase serves as an input for waveform creating blocks, that transform it into amplitude using either a look-up table or internal logic. Finally, a PWM output is created.

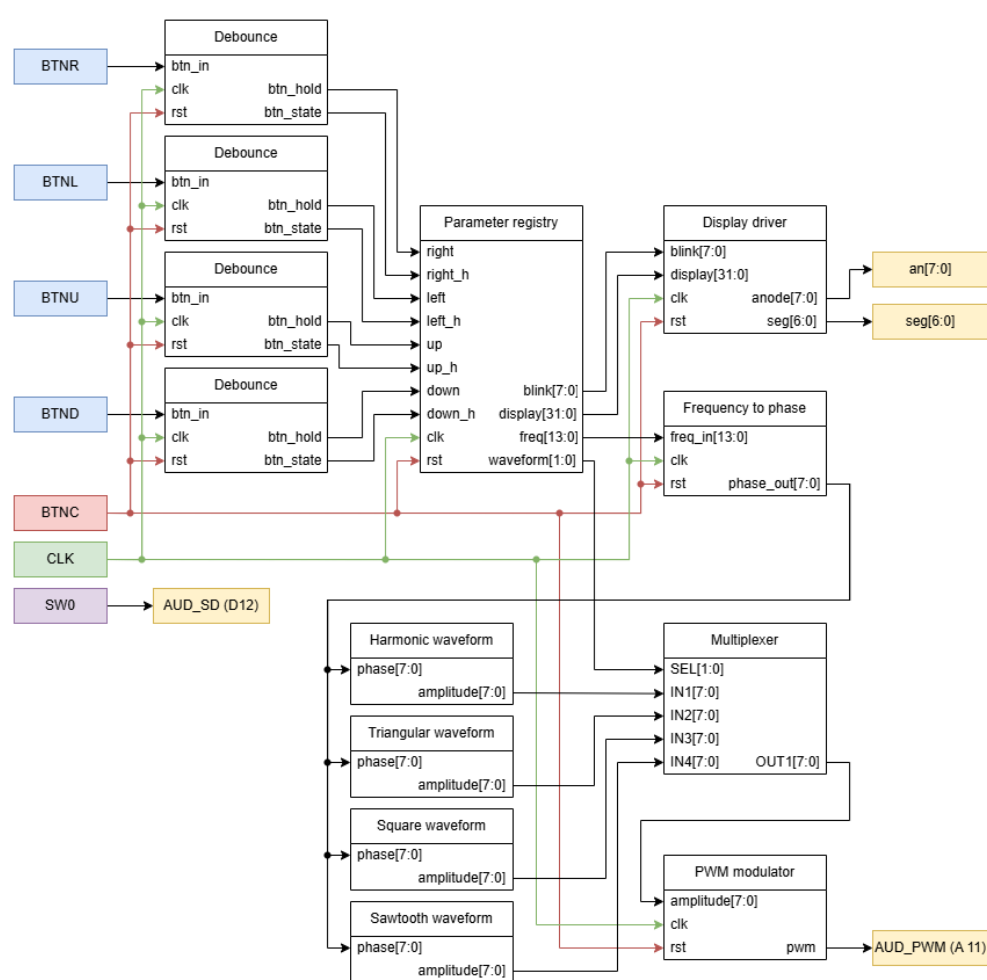
Inputs:

Buttons:

BTNR – change of parameter to modify (shift to the right)
BTNL – change of parameter to modify (shift to the left)
BTNU – incrementation of the selected parameter
BTND – reduction of the selected parameter
BTNC – reset into default state

Switches:

SW0 – switch for enabling output



Video demonstration



GitHub page